EXPT. NO. : 1

Aim:

Introduction to STM32 Development Board and Installation of STM32 Cube IDE.

Objectives:

Installation of Cube IDE for code writing and identification of components on STM32 Development board.

Introduction

We are very much thankful to you for procuring STM32 Microcontroller Development Board - "**Explore-STM32-V1.0"**.We have introduced an innovative design tosupport all kinds of Nucleo-64 STM32 boards from ST Microcontrollers by means of uniqueconceptof mother boardand daughterboard.

The mother board consists of all the peripherals required to design most of the embedded systems.The Nucloe-64 board will act as daughter board which consists of minimal circuit of microcontroller board, power supply, port connector and connector tocommunicate with base board. One of the innovative features of daughter board is that it can be used independently as mini development board.

Features

- Supports all kinds of Nucleo-64 Boards
    - STM32F0SeriesNucleo-64Board
    - STM32F1SeriesNucleo-64Board
    - STM32F3SeriesNucleo-64Board
    - STM32F4SeriesNucleo-64Board
- In-system programming facility
- USB Programmable(USB Bootloader)
- RS232(UART) Interface
- 6-UserProgrammable LEDs
- 16x 2 LCD Interface I2C based EEPROM(AT24Cxx) interface
- SPI based EEPROM (AT25Cxx),
- Analog input interface–LM35,LD Rand Trimpot
- 4-Push buttons (also act as External Interrupt)
- Buzzer Interface and Relay Interface
- DC Motor Driver (L293D)Interface and Stepper Motor Interface
- RGBLED Interface
- 2-DigitMultiplexedSevenSegmentDisplay

- Provision for interfacing Peripheral Boards
    - Sensor interface board

- TFT Interface board
- LoRA Interface board

- All ports open for peripheral connections
- User selectable port pins for peripheral interface

Getting started with Board

1. **PowerSection**

   The board can power using 9VDC adapter. This section generates the 5V and 3.3V which required for Nucleo-64 board and other peripherals on the board

2. **PushButtonSection**

   This section consists of 4 push buttons which can be used for providing digitalinput to microcontroller or act as interrupt source. The 4-way DIP switch is used tointerface these push buttons are connected to certain microcontroller port pin. When DIPswitch is in ON position; push buttons connected to specified port pin of microcontrollerotherwise user can connect to any other port pin of microcontroller by putting DIP switchinOFFposition.

PB_SW	JP7	Designation	STM32PortPin
1–ON	1	SW1	PB0
2–ON	2	SW2	PB1
3–ON	3	SW3	PB5
4–ON	4	SW4	PA15

3. **I2C and SPI Section**

This section consists of **I2C EEPROM–AT24C04** and SPI**EEPROM–AT25C128/256**. The 6-way DIP switch is used to interface I2C devices to I2C port pinof microcontroller and SPI Devices to SPI port pins of microcontroller. When DIP switch is in ON position; I2C/SPI devices connected to I2C-1 and SPI-1port pin of microcontroller otherwise user can connect to I2C-2 or SPI- 2port pin of microcontroller by putting DIP switch in OFF position.

SCI_SW	JP11	Signal	STM32PortPin
1-ON	1	SCL	PB8(I2C1_SCL)
2-ON	2	SDA	PB9(I2C_SDA)
3-ON	3	SCK	PA5(SPI1_CLK)
4-ON	4	SDI	PA6(SPI1_MISO)
5-ON	5	SDO	PA7(SPI1_MOSI)
6-ON	6	CS_EEPROM	PB7

4. **DCMotorSection**

   This section consists of Driver IC L293D which is used to either drive steppermotor or DC Motor. The connection details are as shown below. The 6-way DIP switch is used to interface L239D to particular pins of Microcontroller.When DIP switch is in ON position; L293D connected to specified port pin of microcontroller other wise

user can connect to any other port pin of microcontroller by putting DIP switch in OFF position.

DCM_SW	JP8	Signal	STM32PortPin
1-ON	1	EN1	PB3(TIM2_CH2)
2-ON	2	IN1	PB12
3-ON	3	IN2	PB13
4-ON	4	EN2	PB10(TIM2_CH3)
5-ON	5	IN3	PB9
6-ON	6	IN4	PB8

5. LED Section

This section consists of 6 LED's for digital output. LED's are connected in Common Cathode Mode. The connection details are as shown below. The 8-way DIP switch is used to interface LED's to particular pins of Microcontroller. When DIP switch is in ON position; LED's connected to specified port pin of microcontroller other wise user can connect to any other port pin of microcontroller by putting DIP switch in OFF position.

LED_SW	JP6	Signal	STM32PortPin
1-ON	1	L0	PC8
2-ON	2	L1	PC9
3-ON	3	L2	PC10
4-ON	4	L3	PC11
5-ON	5	L4	PC12
6-ON	6	L5	PC13

6. RGBLED/ Buzzer/Relay Section

This section consists of buzzer for audio indication, relay to control the appliances and RGB LED. The 6-Way DIP switch is used to interface these inputs to Microcontroller Pins. When DIP switch is in ON position; peripherals connected tospecified port pin of microcontroller otherwise user can connect to any other port pin of microcontroller by putting DIP switch in OFFposition.

PERI_SW	JP13	Signal	STM32PortPin
1-ON	1	RGB_Red	PA8/TIM1_CH1
2-ON	2	RGB_Green	PA9/TIM1_CH2
3-ON	3	RGB_Blue	PA10/TIM1_CH3
4-ON	4	Buzzer	PB4/TIM3_CH1
5-ON	5	Relay	PD2
6-ON	6	NC	NC

7. LCD 16x2 Section

This section consists for 16x2 LCD interfacing provision. The DIP switch is usedto interface these inputs to Microcontroller Pins. When DIP switch is in ON position;peripherals connected to specified port pin of microcontroller otherwise user can connect to any other port pin of microcontroller by putting DIP switch in OFF position.

LCD_SW1	JP1	Signal	STM32PortPin
1-ON	1	LCD_RS	PC8
2-ON	2	LCD_EN	PC9

LCD_SW2	JP1	Signal	STM32PortPin
1-ON	3	LCD_D0	PC0
2-ON	4	LCD_D1	PC1
3-ON	5	LCD_D2	PC2
4-ON	6	LCD_D3	PC3
5-ON	7	LCD_D4	PC4
6-ON	8	LCD_D5	PC5
7-ON	9	LCD_D6	PC6
8-ON	10	LCD_D7	PC7

8. Multiplexed 7-Segment Display

This section consists for two 7-segment display interfaced in multiplexed fashion. The 7-segment displays are in common anode fashion. When DIP switch is in ON position;
Peripherals connected to specified port pin of microcontroller otherwise user can connect to any other port pin of microcontroller by putting DIP switch in OFF position.

SEG_SW1	JP2	Signal	STM32PortPin
1-ON	1	DISP1	PB14
2-ON	2	DISP2	PB15

SEG_SW2	JP2	Signal	STM32PortPin
1-ON	3	SEG_A	PC0
2-ON	4	SEG_B	PC1
3-ON	5	SEG_C	PC2
4-ON	6	SEG_D	PC3
5-ON	7	SEG_E	PC4
6-ON	8	SEG_F	PC5
7-ON	9	SEG_G	PC6
8-ON	10	SEG_DP	PC7

9. Analog Input Interface Section

This section consists of 4 Analog inputs wiz two variable potentiometer,

TemperatureSensor LM35 and LDR. The DIP switch is used to interface these analog inputs to AnalogPins of Microcontroller. When DIP switch is open then Analog Inputs will be connectedSpecifiedAnalogPins otherwise user canconnectto anyother analoginput pin.

ADC_SW	JP6	Signal	STM32 Port Pin
1-ON	1	LM35	PA0(ADC123_IN0)
2-ON	2	AIN1	PA1(ADC123_IN1)
3-ON	3	LDR	PA4(ADC12_IN4)
4-ON	4	AIN2	PB1(ADC12_IN9)

10.Serial Communication Interface

This section consists of RS232 serial interface. By default UART3 is connected toMAX232whenpins of 'UART TTL' areconnected.

UART_TTL Connector		
Pin No.	**Pin Function**	**STM32 Port Pin**
1	5V	5V
2	TXD	PC10/USART3_TX
3	RXD	PC11/USART3_RX
4	GND	GND

11.CAN (Controller Area Network) Interface

This section consists of CAN serial interface. By default CAN 1isused.

CAN_SW	Signal	STM32PortPin
1-ON	LM35	PA12/CAN1_TX
2-ON	AIN1	PA11/CAN1_RX

Peripheral Connection at a Glance

1. **Push Button Section**

PB_SW	JP7	Designation	STM32PortPin
1–ON	1	SW1	PB0
2–ON	2	SW2	PB1
3–ON	3	SW3	PB5
4–ON	4	SW4	PA15

2. **I2C and SPI Section**

SCI_SW	JP11	Signal	STM32PortPin
1-ON	1	SCL	PB8(I2C1_SCL)
2-ON	2	SDA	PB9(I2C_SDA)
3-ON	3	SCK	PA5(SPI1_CLK)
4-ON	4	SDI	PA6(SPI1_MISO)

5-ON	5	SDO	PA7(SPI1_MOSI)
6-ON	6	CS_EEPROM	PB7

3. DC Motor Section

DCM_SW	JP8	Signal	STM32PortPin
1-ON	1	EN1	PB3(TIM2_CH2)
2-ON	2	IN1	PB12
3-ON	3	IN2	PB13
4-ON	4	EN2	PB10(TIM2_CH3)
5-ON	5	IN3	PB9
6-ON	6	IN4	PB8

4. LED Section

LED_SW	JP6	Signal	STM32PortPin
1-ON	1	L0	PC8
2-ON	2	L1	PC9
3-ON	3	L2	PC10
4-ON	4	L3	PC11
5-ON	5	L4	PC12
6-ON	6	L5	PC13

5. RGBLED/ Buzzer/Relay Section

PERI_SW	JP13	Signal	STM32PortPin
1-ON	1	RGB_Red	PA8/TIM1_CH1
2-ON	2	RGB_Green	PA9/TIM1_CH2
3-ON	3	RGB_Blue	PA10/TIM1_CH3
4-ON	4	Buzzer	PB4/TIM3_CH1
5-ON	5	Relay	PD2
6-ON	6	NC	NC

6. LCD 16x2 Section

LCD_SW1	JP1	Signal	STM32PortPin
1-ON	1	LCD_RS	PC8
2-ON	2	LCD_EN	PC9

LCD_SW2	JP1	Signal	STM32PortPin
1-ON	3	LCD_D0	PC0
2-ON	4	LCD_D1	PC1
3-ON	5	LCD_D2	PC2
4-ON	6	LCD_D3	PC3

5-ON	7	LCD_D4	PC4
6-ON	8	LCD_D5	PC5
7-ON	9	LCD_D6	PC6
8-ON	10	LCD_D7	PC7

7. Multiplexed 7-Segment Display

SEG_SW1	JP2	Signal	STM32PortPin
1-ON	1	DISP1	PB14
2-ON	2	DISP2	PB15

SEG_SW2	JP2	Signal	STM32PortPin
1-ON	3	SEG_A	PC0
2-ON	4	SEG_B	PC1
3-ON	5	SEG_C	PC2
4-ON	6	SEG_D	PC3
5-ON	7	SEG_E	PC4
6-ON	8	SEG_F	PC5
7-ON	9	SEG_G	PC6
8-ON	10	SEG_DP	PC7

8. Analog Input Interface Section

ADC_SW	JP6	Signal	STM32PortPin
1-ON	1	LM35	PA0(ADC123_IN0)
2-ON	2	AIN1	PA1(ADC123_IN1)
3-ON	3	LDR	PA4(ADC12_IN4)
4-ON	4	AIN2	PB1(ADC12_IN9)

9. Serial Communication Interface

UART_TTLConnector		
PinNo.	**PinFunction**	**STM32PortPin**
1	5V	5V
2	TXD	PC10/USART3_TX
3	RXD	PC11/USART3_RX
4	GND	GND

10. CAN(ControllerAreaNetwork) Interface

CAN_SW	Signal	STM32PortPin
1-ON	LM35	PA12/CAN1_TX
2-ON	AIN1	PA11/CAN1_RX

Troubleshooting

Name of the Section	Component Description	Probable Repair
Power Supply Section	1.LM7805(U1) 2.LM1117– 3V3 (PU2)	1.Check Power Jack
I2C Section	1.I2CEEPROM–AT24C04(U3)	1. DIP Switch position should be ON while performing I2C Experiments 2. Replace IC
SPI Section	1.SPI EEPROM – AT25C256(U6)	1. DIP Switch position should be on while performing SPI expts. 2. Replace IC's if not working
DC Motor Section	1.L239D (U7)	1. 6-way DIP Switch position should be on while performing practical. 2. External power must be given for DC motor interfacing
LED Array Section	Red LED 3mm 470E Resistor Array– 9pin	1. Check each LED by applying 5V to corresponding LED pin 2. If not glowing Replace LED
LCD Section	16x2LCD,10Kpot(3386)	1. Adjust contrast such that block squares must be displayed 2. If Contrast is not adjusted properly,check voltage at pin no. 3 of LCD, itshould belessthan 1V 3. Check 10k pot it may open, if voltageisnot varying.
RGB/ Relay/ Buzzer	1. Common Cathode RGBLED 2. 5V Sugar cubeRelay 3. 5Vbuzzer PCB mounted	1. RGB LED may be open if high voltage is provided. 2. Give direct 5V supply to relay coil to test relay. 3. Give direct 5V supply to buzzer to check the buzzer
Serial Interface	MAX232, 10uF/16V Capacitor,	1. If IC is overheating, replace the IC. 2. Remove the daughter board and short RX1 and TX1 to check the loop back Communication.

Conclusion: _____
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